Lewis Smith. The 'strong' feature hypothesis could be wrong, Aug 2024. URL https://www.alignmentforum.org/posts/tojtPCCRpKLSHBdpn/the-strong-feature-hypothesis-could-be-wrong.

Adly Templeton, Tom Conerly, Jonathan Marcus, Jack Lindsey, Trenton Bricken, Brian Chen, Adam Pearce, Craig Citro, Emmanuel Ameisen, Andy Jones, Hoagy Cunningham, Nicholas L Turner, Callum McDougall, Monte MacDiarmid, C. Daniel Freeman, Theodore R. Sumers, Edward Rees, Joshua Batson, Adam Jermyn, Shan Carter, Chris Olah, and Tom Henighan. Scaling monosemanticity: Extracting interpretable features from claude 3 sonnet. *Transformer Circuits Thread*, 2024. URL https://transformer-circuits.pub/2024/scaling-monosemanticity/index.html.

A Analyzing different perturbations setups and synthetic activations

In the main paper we use absolute step size for perturbations, however blowup locations have a dependence on the distance between the base and target activations, which can make the MS step distributions with absolute step size misleading. We know that the blowup location does not solely depend on the distance between the base and target activations, and in order to isolate this property, we create a distance agnostic setup using relative step sizes. In the relative step size approach, our perturbations always start at a base activation (A) and end at a target activation (T) using linear interpolation:

$$\mathrm{A}_{\mathrm{pert}}(\mathrm{n}) = \left(1 - \frac{\mathrm{n}}{100}\right) \cdot \mathrm{A} + \frac{\mathrm{n}}{100} \cdot \mathrm{T}$$

where n is the perturbation step, which goes from 0 to 100. This method ensures that we always transition from the base activation to the target activation in a fixed number of steps, regardless of the distance between them. By using relative step size, we remove the dependence of the blowup location on distance, and instead compare the effect of perturbations purely in terms of the percentage of base and target activations present at each step. For example, step 50 in this setup implies that the perturbed activation is made up of 50% base activation and 50% target activation.

In the relative step size setup, we find that the MS step distribution for perturbations towards model-generated activations peaks more strongly around step 50 than in the absolute step size setup. The blowups are also localized between step 30 and 70, implying that blowups usually happen in the middle of the perturbation (Figure A.1). We posit that until step 30, the model treats the interpolated activation as the base activation. This is due to 70% of the interpolated activation coming from the base activation, and the remaining 30% coming from the target activation being treated as noise. This effect reverses at step 70, where the model starts treating the interpolated activation as the target activation, and the 30% that comes from the base activation is considered noise.

Our analysis reveals that the MS step distribution for random activation perturbations exhibits marginally higher variance than the absolute step size setup, with a rightward shift relative to the distribution for model-generated activation perturbations (Table A.1). This suggests that stronger perturbations towards random activations are required to induce a blowup compared to model-generated activations. Furthermore, it indicates that the model is more resilient to random noise than to noise directed towards another model-generated activation, requiring a greater magnitude of the former to cause confusion in the model.

In this setup, comparing perturbations with synthetic-baseline and synthetic-structured activations reveals that while synthetic-structured activations still more closely mimic model-generated activations, the disparity between the two has notably decreased (Figure A.1, Table A.1). This suggests that synthetic-baseline activations less effectively align with the residual stream geometry of model-generated activations compared to synthetic-structured ones, explaining the latter's superior performance in the absolute step size setup. Our findings indicate that considering latent sparsity is important for synthetic activations to emulate model-generated activations in the relative step size setup. Consequently, both synthetic-structured and synthetic-baseline outperform synthetic activations created using the "bag of SAE latents" approach without accounting for sparsity (synthetic-random).

We find that when we construct synthetic-structured activations (Section 4.3), omitting the cosine similarity constraint on the top latent and instead selecting based on sparsity similarity to the base activation's top latent yields the best-performing synthetic activations in the relative step size setup.

However, these activations typically have greater distance from the base activation compared to synthetic-structured activations. Consequently, their performance in the absolute step size scenario is inferior to that of synthetic-structured activations.

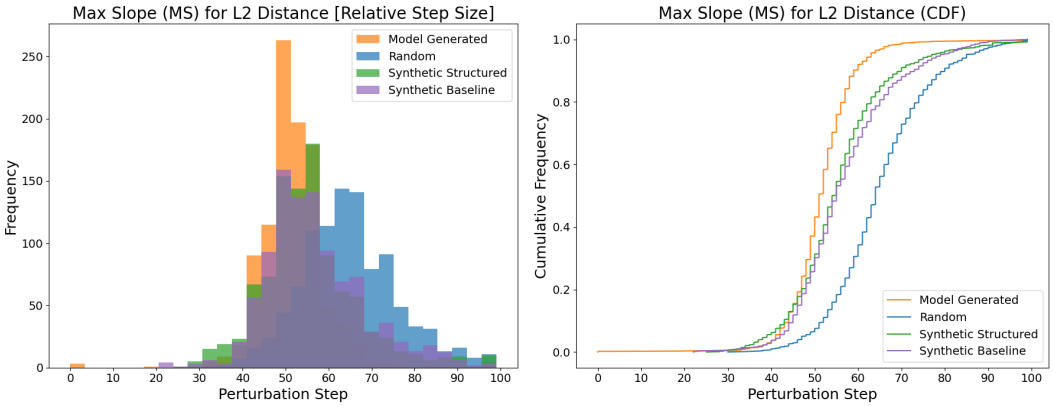

Figure A.1: The distributions of the max slope (MS) steps for perturbations with relative step size towards model-generated (orange), random (blue), synthetic-baseline (purple), and synthetic-structured (green) activations. The left panel shows the counts of MS steps occurring in different bins along the length of the perturbation, and the right panel shows the cumulative frequency for the same. We find that perturbing towards synthetic-structured activations in the relative step size setup is slightly more similar to perturbing towards model-generated activations than perturbing towards synthetic-baseline activations is.

Max Slope (MS) Step Distribution Statistics

Activation Type	Absolute Step Size			Relative Step Size		
	Mean	Std dev	KS	Mean	Std dev	KS
Model Generated	41.21	10.32	0.00	51.65	7.42	0.00
Random	52.49	10.34	0.44	64.89	10.50	0.62
Synthetic Baseline	49.88	12.60	0.31	57.07	11.29	0.27
Synthetic Structured	**43.45**	12.78	**0.11**	55.69	11.31	0.22
Synthetic Random	51.30	**10.25**	0.39	55.25	**8.74**	0.19
Synthetic Structured (w/o cos sim)	50.17	11.96	0.31	**54.47**	10.68	**0.17**

Table A.1: We find that controlling for the sparsity of the top latent and the cosine similarity between the active latents play an important role in making synthetic-structured activations perform well in both absolute and relative step setups. This table contains the mean, standard deviation and KS statistic for MS step distributions for all types of synthetic activations we tested. The KS statistic is measured against perturbations towards model-generated activations, with a lower value meaning higher similarity. The entries in bold show the best match with statistics for model-generated activations.

B Metrics for analysing blowups

In our main analysis, we focus on the maximum slope (MS) as an indicator of the blowup step. In this section we share findings using the Area Under the Curve (AUC) and Non Linear (NL) metrics to represent important parts of the L2 distance vs perturbation step curve.

B.1 Area Under Curve (AUC)

Our experimental results reveal that certain L2 distance curves deviate from the expected step-function-like pattern, causing the MS step to misrepresent the actual blowup location for these curves. In contrast, the AUC metric provides a more comprehensive assessment of activation behavior across the entire perturbation process. This approach not only identifies the steepest increase point but also effectively screens out atypical curves that might otherwise evade detection. AUC calculates the step at which the following ratio is maximized:

$$R = \text{area of the triangle defined by (0,0), (x,0) and (f(x),x)}/\text{area under the curve f(x)}$$

where $f(x)$ is L2 distance as a function of the perturbation step x. This method is sensitive to the concavity or convexity of the perturbation curve. For predominantly concave curves (where the rate of change increases over time), the AUC blowup step tends to occur later, as the triangular area takes longer to outpace the actual area under the curve. Conversely, for convex curves (where the rate of change decreases over time), the AUC blowup step tends to occur earlier. This property allows the AUC method to implicitly capture information about the shape of the perturbation.

The AUC metric serves as sanity check, confirming that most perturbations align with expectations. Convex L2 distance curves yield early AUC peaks, and Figure B.1 demonstrates that the majority of perturbations exhibit the anticipated concave shape. We find that our perturbation results hold for AUC step distributions in the absolute step size setup (Table B.1), with structured-synthetic activations more closely mimicking model-generated activations compared to synthetic-baseline activations. In the relative step size setup (detailed in Appendix A), synthetic-structured and synthetic-baseline activations perform similarly. This can be attributed to the higher prevalence of convex curves in perturbations towards synthetic-structured activations versus synthetic-baseline activations.

Area Under Curve (AUC) Step Distribution Statistics

Activation Type	Absolute Step Size			Relative Step Size		
	Mean	Std dev	KS	Mean	Std dev	KS
Model Generated	41.94	11.78	0.00	51.98	9.64	0.00
Random	52.73	13.66	0.43	64.97	16.01	0.59
Synthetic Baseline	49.31	14.20	0.25	56.66	13.20	0.22
Synthetic Structured	43.54	14.99	0.09	54.84	15.51	0.21

Table B.1: We find that our results for the AUC step distributions are similar to those for the MS step distributions. This table contains the mean, standard deviation and KS statistic for AUC step distributions for all the perturbations we perform. The KS statistic is measured against perturbations towards model-generated activations, with a lower value meaning higher similarity.